

Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

- 1 Output action and equivariance
- 2 Lift of the VSLAM kinematics
- 3 Observer construction

Visual SLAM problem

Visual simultaneous localisation and mapping (VSLAM):

A moving camera observes fixed landmarks and estimates simultaneously:

- the robot pose,
- the landmark positions,
- possibly unknown depths from bearing-only measurements.

Robot pose:

$$P = \begin{pmatrix} R_P & x_P \\ 0 & 1 \end{pmatrix} \in \text{SE}(3), \quad p_i \in \mathbb{R}^3, \quad i = 1, \dots, n.$$

Raw SLAM coordinates:

$$(P, p_1, \dots, p_n) \in \text{SE}(3) \times (\mathbb{R}^3)^n.$$

Rigid-body velocity input

The measured rigid-body velocity is written as

$$U = \begin{pmatrix} \Omega_U^\times & V_U \\ 0 & 0 \end{pmatrix} \in \mathfrak{se}(3),$$

where

$$\Omega_U \in \mathbb{R}^3, \quad V_U \in \mathbb{R}^3.$$

The kinematics are

$$\dot{P} = PU, \quad \dot{p}_i = 0.$$

Equivalently,

$$\frac{d}{dt}(P, p_i) = f((P, p_i), U) := (PU, 0).$$

Bearing-only camera output

The camera measures only directions to landmarks:

$$y_i = h^i(P, p_i) := \frac{R_P^\top (p_i - x_P)}{\|p_i - x_P\|} \in \mathbb{S}^2.$$

Define the ego-centric landmark coordinate

$$q_i := R_P^\top (p_i - x_P).$$

Then

$$y_i = \frac{q_i}{\|q_i\|}, \quad q_i = r_i y_i, \quad r_i := \|p_i - x_P\|.$$

Key observation

The measurement contains direction but not range.

Why VSLAM is difficult

① Nonlinear outputs

$$y_i = \frac{R_P^\top (p_i - x_P)}{\|p_i - x_P\|}.$$

② Gauge freedom

Absolute inertial coordinates are arbitrary.

③ Depth ambiguity

A single bearing measurement does not determine

$$r_i = \|p_i - x_P\|.$$

④ High dimension

The raw coordinate dimension is

$$6 + 3n.$$

SLAM manifold and gauge symmetry

Let

$$S = \begin{pmatrix} R_S & x_S \\ 0 & 1 \end{pmatrix} \in \text{SE}(3)$$

be a change of inertial frame.

The same physical configuration is represented by

$$(P, p_i) \sim (S^{-1}P, R_S^\top(p_i - x_S)).$$

Interpretation

Changing the world frame changes coordinates but not the physical SLAM configuration.

Reduced total space

Bearings are well-defined only when the robot and landmarks are not collocated:

$$p_i \neq x_P, \quad i = 1, \dots, n.$$

Define

$$\mathcal{T}_n^\circ(3) = \left\{ (P, p_i) \in \text{SE}(3) \times (\mathbb{R}^3)^n \mid p_i \neq x_P, i = 1, \dots, n \right\}.$$

The trajectory is assumed to remain in $\mathcal{T}_n^\circ(3)$.

Equivalence classes

Define

$$[P, p_i] := \{ (S^{-1}P, R_S^\top(p_i - x_S)) \mid S \in \text{SE}(3) \}.$$

The VSLAM manifold is the quotient

$$\mathcal{M}_n^\circ(3) := \{ [P, p_i] \mid (P, p_i) \in \mathcal{T}_n^\circ(3) \}.$$

Important

The physical state is not a particular coordinate tuple (P, p_i) , but an equivalence class $[P, p_i]$.

Ego-centric coordinates are gauge invariant

For a configuration (P, p_i) , define

$$q_i = R_P^\top (p_i - x_P).$$

For the transformed coordinates

$$P' = S^{-1}P, \quad p'_i = R_S^\top (p_i - x_S),$$

we have

$$R_{P'} = R_S^\top R_P, \quad x_{P'} = R_S^\top (x_P - x_S).$$

Then

$$\begin{aligned} R_{P'}^\top (p'_i - x_{P'}) &= (R_S^\top R_P)^\top (R_S^\top (p_i - x_S) - R_S^\top (x_P - x_S)) \\ &= R_P^\top R_S R_S^\top (p_i - x_P) \\ &= R_P^\top (p_i - x_P) \\ &= q_i. \end{aligned}$$

Observable information

Since

$$q_i = R_P^\top (p_i - x_P)$$

is invariant under gauge transformations, the bearing output

$$y_i = \frac{q_i}{\|q_i\|}$$

is also well-defined on the quotient manifold.

Two configurations are equivalent on the SLAM manifold iff

$$R_{P1}^\top (p_i^1 - x_{P1}) = R_{P2}^\top (p_i^2 - x_{P2})$$

for all i .

Core idea

The SLAM manifold can be understood through ego-centric landmark coordinates.

A new VSLAM symmetry group: why a new symmetry?

For equivariant observer design we seek group actions satisfying

$$h(\Phi(X, \xi)) = \rho(X, h(\xi)).$$

For full landmark positions, classical SLAM symmetries are natural.

For bearing-only outputs, classical symmetries are not sufficient.

Goal

Construct a symmetry group whose action is compatible with bearing measurements.

The VSLAM group

Define

$$\text{VSLAM}_n(3) = \text{SE}(3) \times (\text{SO}(3) \times \mathbb{R}_+)^n.$$

An element is written

$$X = (A, (Q, a)_i) = (A, (Q_1, a_1), \dots, (Q_n, a_n)).$$

where

$$A \in \text{SE}(3), \quad Q_i \in \text{SO}(3), \quad a_i \in \mathbb{R}_+.$$

Dimension:

$$\dim \text{VSLAM}_n(3) = 6 + n(3 + 1) = 6 + 4n.$$

Group product, identity, inverse

Group product:

$$(A_1, (Q_1, a_1)_i) \cdot (A_2, (Q_2, a_2)_i) = (A_1 A_2, (Q_1 Q_2, a_1 a_2)_i).$$

Identity:

$$\text{id} = (I_4, (I_3, 1)_i).$$

Inverse:

$$(A, (Q, a)_i)^{-1} = (A^{-1}, (Q^\top, a^{-1})_i).$$

State action on the total space

Define

$$\Phi : \text{VSLAM}_n(3) \times \mathcal{T}_n^\circ(3) \rightarrow \mathcal{T}_n^\circ(3)$$

by

$$\Phi((A, (Q, a)_i), (P, p_i)) = (PA, a_i^{-1} R_{PA} Q_i^\top R_P^\top (p_i - x_P) + x_{PA}).$$

The pose is transformed as

$$P \mapsto PA.$$

The landmark is transformed through its body-fixed coordinate.

Geometric meaning of the state action

For each landmark:

$$q_i = R_P^\top (p_i - x_P).$$

The action performs:

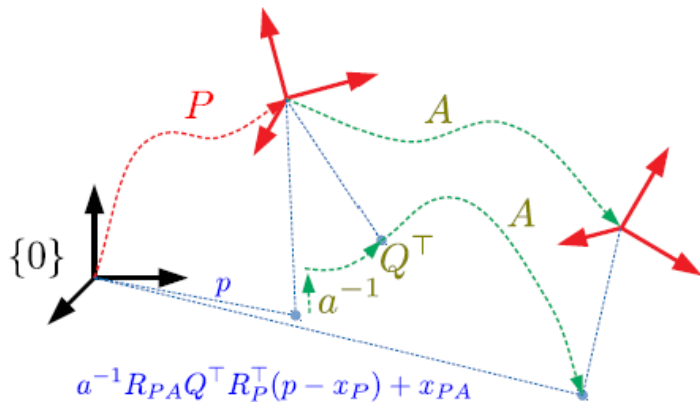
$$q_i \mapsto a_i^{-1} Q_i^\top q_i.$$

Then the transformed landmark is returned to inertial coordinates associated with PA :

$$p'_i = R_{PA}(a_i^{-1} Q_i^\top q_i) + x_{PA}.$$

Interpretation

The group independently rotates and scales the ego-centric coordinate of each landmark.



Proof that Φ is a right action: identity

We check the identity element:

$$\text{id} = (I_4, (I_3, 1)_i).$$

Then

$$\begin{aligned}\Phi(\text{id}, (P, p_i)) &= (PI_4, 1^{-1}R_P I_3^\top R_P^\top (p_i - x_P) + x_P) \\ &= (P, p_i).\end{aligned}$$

So the identity axiom holds.

Proof that Φ is a right action: composition

Let

$$X_1 = (A_1, (Q_1, a_1)_i), \quad X_2 = (A_2, (Q_2, a_2)_i).$$

For a right action, we need

$$\Phi(X_1, \Phi(X_2, \xi)) = \Phi(X_2 X_1, \xi).$$

Applying X_2 first gives

$$P' = PA_2, \quad p'_i = a_2^{-1} R_{PA_2} Q_2^\top R_P^\top (p_i - x_P) + x_{PA_2}.$$

Then applying X_1 gives

$$P'' = PA_2 A_1.$$

Composition computation

The body-fixed coordinate after the first transformation is

$$R_{P'}^\top(p'_i - x_{P'}) = a_2^{-1} Q_2^\top R_P^\top(p_i - x_P).$$

Applying X_1 :

$$\begin{aligned} p''_i &= a_1^{-1} R_{PA_2A_1} Q_1^\top R_{P'}^\top(p'_i - x_{P'}) + x_{PA_2A_1} \\ &= a_1^{-1} a_2^{-1} R_{PA_2A_1} Q_1^\top Q_2^\top R_P^\top(p_i - x_P) + x_{PA_2A_1} \\ &= (a_2 a_1)^{-1} R_{PA_2A_1} (Q_2 Q_1)^\top R_P^\top(p_i - x_P) + x_{PA_2A_1}. \end{aligned}$$

Thus

$$\Phi(X_1, \Phi(X_2, \xi)) = \Phi(X_2 X_1, \xi).$$

Transitivity of the action

Given arbitrary

$$(P, p_i), \quad (P', p'_i)$$

in $\mathcal{T}_n^\circ(3)$, we construct $X = (A, (Q, a)_i)$ such that

$$\Phi(X, (P, p_i)) = (P', p'_i).$$

Choose

$$A = P^{-1}P'.$$

Then

$$PA = P'.$$

For each landmark choose

$$a_i = \frac{\|p_i - x_P\|}{\|p'_i - x_{P'}\|}.$$

Choosing Q_i

We need

$$a_i^{-1} R_{P'} Q_i^\top R_P^\top (p_i - x_P) + x_{P'} = p'_i.$$

Equivalently,

$$Q_i^\top \frac{R_P^\top (p_i - x_P)}{\|p_i - x_P\|} = \frac{R_{P'}^\top (p'_i - x_{P'})}{\|p'_i - x_{P'}\|}.$$

This is possible because $Q_i \in \text{SO}(3)$ can map any unit vector to any other unit vector.

Thus Φ is transitive.

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Output space

Each bearing belongs to the sphere:

$$y_i \in \mathbb{S}^2.$$

The full output space is

$$\mathcal{N}^n(3) = \mathbb{S}^2 \times \dots \times \mathbb{S}^2.$$

The combined output map is

$$h(P, p_i) = \left(\frac{R_P^\top (p_1 - x_P)}{\|p_1 - x_P\|}, \dots, \frac{R_P^\top (p_n - x_P)}{\|p_n - x_P\|} \right).$$

Output action

Define

$$\rho : \text{VSLAM}_n(3) \times \mathcal{N}^n(3) \rightarrow \mathcal{N}^n(3)$$

by

$$\rho((A, (Q, a)_i), y_i) = Q_i^\top y_i.$$

Why does a_i not appear?

Bearings live on $\mathbb{S}^2$, so scaling does not affect the measured direction.

Equivariance theorem

Theorem

The output map h is equivariant with respect to Φ and ρ :

$$h(\Phi(X, \xi)) = \rho(X, h(\xi)).$$

This is the central geometric property that enables an intrinsic output error.

Equivariance proof

Let

$$X = (A, (Q, a)_i), \quad \xi = (P, p_i).$$

Then

$$\Phi(X, \xi) = (PA, a_i^{-1} R_{PA} Q_i^\top R_P^\top (p_i - x_P) + x_{PA}).$$

Therefore

$$\begin{aligned} h^i(\Phi(X, \xi)) &= \frac{R_{PA}^\top (a_i^{-1} R_{PA} Q_i^\top R_P^\top (p_i - x_P))}{\|a_i^{-1} R_{PA} Q_i^\top R_P^\top (p_i - x_P)\|} \\ &= \frac{a_i^{-1} Q_i^\top R_P^\top (p_i - x_P)}{a_i^{-1} \|p_i - x_P\|} \\ &= Q_i^\top h^i(P, p_i) \\ &= \rho^i(X, h^i(\xi)). \end{aligned}$$

Why equivariance matters

Equivariance gives a natural output error:

$$d_i = \rho(\hat{X}^{-1}, y_i).$$

This error is intrinsic because it is defined using the group action, not by subtracting local coordinates.

At the desired equilibrium,

$$d_i = y_i^\circ,$$

where

$$y_i^\circ = h^i(\xi^\circ)$$

is the bearing associated with the chosen origin configuration.

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Why lift the system?

Original dynamics live on the total space:

$$\dot{\xi} = f(\xi, U).$$

We want observer dynamics on the Lie group $\text{VSLAM}_n(3)$.

A lift is a map

$$\Lambda : \mathcal{T}_n^\circ(3) \times \mathfrak{se}(3) \rightarrow \mathfrak{vslam}_n(3)$$

such that

$$D\Phi_{(P, p_i)}(\text{id})[\Lambda((P, p_i), U)] = f((P, p_i), U).$$

Definition of the lift

Let

$$q_i = R_P^\top (p_i - x_P).$$

Define

$$\Lambda((P, p_i), U) = (U, (\Lambda_Q(U, q_i), \Lambda_a(U, q_i))_i).$$

where

$$\Lambda_Q(U, q) = \left(\Omega_U + \frac{q \times V_U}{\|q\|^2} \right)^\times$$

and

$$\Lambda_a(U, q) = \frac{q^\top V_U}{\|q\|^2}.$$

Interpretation of Λ_Q

$$\Lambda_Q(U, q) = \left(\Omega_U + \frac{q \times V_U}{\|q\|^2} \right)^\times.$$

The term Ω_U accounts for body rotation.

The term

$$\frac{q \times V_U}{\|q\|^2}$$

accounts for bearing rotation induced by translational velocity transverse to the line of sight.

If $V_U \parallel q$, then

$$q \times V_U = 0,$$

so there is no bearing rotation due to translation.

Interpretation of Λ_a

$$\Lambda_a(U, q) = \frac{q^\top V_U}{\|q\|^2}.$$

This measures the radial component of translational velocity along the landmark direction.

If $V_U \perp q$, then

$$q^\top V_U = 0,$$

and the depth does not change instantaneously.

If $V_U \parallel q$, then the range changes but the bearing does not rotate.

Lift proof: setup

Let

$$W_i^\times = \Lambda_Q(U, q_i), \quad w_i = \Lambda_a(U, q_i).$$

We compute the derivative of the action at the identity:

$$D\Phi_{(P, p_i)}(\text{id})[(U, (W_i^\times, w_i))] = (PU, v_i).$$

The landmark component is

$$v_i = \left. \frac{d}{ds} \right|_{s=0} \left[(1 + sw_i)^{-1} R_{P(l_4 + sU)} (I_3 + sW_i^\times)^\top q_i \right. \\ \left. + x_{P(l_4 + sU)} \right].$$

Lift proof: derivative

Using

$$\left. \frac{d}{ds} \right|_{s=0} (1 + sw_i)^{-1} = -w_i,$$

$$R_{P(I+sU)} = R_P(I + s\Omega_U^\times) + O(s^2),$$

and

$$x_{P(I+sU)} = x_P + sR_P V_U + O(s^2),$$

we obtain

$$v_i = -w_i R_P q_i + R_P \Omega_U^\times q_i + R_P (W_i^\times)^\top q_i + R_P V_U.$$

Since $W_i^\times$ is skew-symmetric,

$$(W_i^\times)^\top = -W_i^\times.$$

Lift proof: substitution

Substitute

$$w_i = \frac{q_i^\top V_U}{\|q_i\|^2}, \quad W_i^\times = \left(\Omega_U + \frac{q_i \times V_U}{\|q_i\|^2} \right)^\times.$$

Then

$$\begin{aligned} v_i = & -\frac{q_i^\top V_U}{\|q_i\|^2} R_P q_i + R_P \Omega_U^\times q_i + R_P V_U \\ & - R_P \left(\Omega_U + \frac{q_i \times V_U}{\|q_i\|^2} \right)^\times q_i. \end{aligned}$$

The Ω_U -terms cancel:

$$R_P \Omega_U^\times q_i - R_P \Omega_U^\times q_i = 0.$$

Lift proof: key vector identity

We use

$$(a \times b)^\times c = (a \times b) \times c.$$

With $a = q$, $b = V$, $c = q$,

$$(q \times V) \times q = V(q^\top q) - q(V^\top q).$$

Therefore

$$\frac{(q \times V)^\times q}{\|q\|^2} = V - \frac{qq^\top}{\|q\|^2} V.$$

Equivalently,

$$(q \times V)^\times q = \|q\|^2 V - qq^\top V.$$

Lift proof: cancellation

Continuing,

$$\begin{aligned} v_i &= -R_P \frac{q_i q_i^\top}{\|q_i\|^2} V_U + R_P V_U - R_P \left(V_U - \frac{q_i q_i^\top}{\|q_i\|^2} V_U \right) \\ &= -R_P \frac{q_i q_i^\top}{\|q_i\|^2} V_U + R_P V_U - R_P V_U + R_P \frac{q_i q_i^\top}{\|q_i\|^2} V_U \\ &= 0. \end{aligned}$$

Hence

$$D\Phi_{(P,p_i)}(\text{id})[\Lambda((P,p_i), U)] = (PU, 0) = f((P,p_i), U).$$

Conclusion

The proposed Λ is a valid lift.

Lifted system

Fix an arbitrary origin configuration

$$\xi^\circ = (P^\circ, p_i^\circ) \in \mathcal{T}_n^\circ(3).$$

The lifted system is

$$\dot{X} = X\Lambda(\Phi(X, \xi^\circ), U).$$

For

$$X = (A, (Q, a)_i),$$

this gives

$$\dot{X} = (AU, (Q\Lambda_Q, a\Lambda_a)_i).$$

If

$$\Phi(X(0), \xi^\circ) = \xi(0),$$

then

$$\xi(t) = \Phi(X(t), \xi^\circ).$$

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Observer state and estimated state

The observer state is

$$\hat{X} = (\hat{A}, (\hat{Q}, \hat{a})_i) \in \text{VSLAM}_n(3).$$

The estimated SLAM configuration is obtained by projection:

$$\hat{\xi} = (\hat{P}, \hat{p}_i) = \Phi(\hat{X}, \xi^o).$$

Important distinction

The observer state $\hat{X}$ lives on the symmetry group. The estimated state $\hat{\xi}$ lives on the VSLAM total space or quotient manifold.

Observer dynamics

The observer is

$$\dot{\hat{X}} = \hat{X} \Lambda(\Phi(\hat{X}, \xi^o), U) - \Delta_{\hat{X}} \hat{X}.$$

where

$$\Delta_{\hat{X}} = (\Delta, (\Gamma, \gamma)_i) \in \mathfrak{vslam}_n(3).$$

Design problem

Choose $\Delta_{\hat{X}}$, especially Γ_i and γ_i , so that landmark bearing and range errors converge.

Origin output

The fixed origin configuration is

$$\xi^\circ = (P^\circ, p_i^\circ).$$

Define the origin bearing

$$y_i^\circ = h^i(\xi^\circ) = \frac{R_{P^\circ}^\top (p_i^\circ - x_{P^\circ})}{\|p_i^\circ - x_{P^\circ}\|}.$$

The target for the intrinsic output error is

$$d_i \rightarrow y_i^\circ.$$

Observer structure

The observer is

$$\dot{\hat{X}} = \hat{X} \Lambda(\Phi(\hat{X}, \dot{\xi}), U) - \Delta_{\hat{X}} \hat{X}.$$

The innovation term is

$$\Delta_{\hat{X}} = (\Delta, (\Gamma_i, \gamma_i)_i).$$

- Γ_i : bearing correction,
- γ_i : range correction,
- barrier term in γ_i : keeps $\hat{r}_i > 0$.

The design objective is

$$d_i \rightarrow \dot{y}_i, \quad \tilde{r}_i \rightarrow 1.$$

Error dynamics

The true range satisfies

$$\dot{r}_i = -d_i^\top \hat{Q}_i V_U.$$

The intrinsic bearing error satisfies

$$\dot{d}_i = \left(\frac{\dot{y}_i^\times \hat{Q}_i V_U}{\hat{r}_i} - \frac{d_i^\times \hat{Q}_i V_U}{r_i} \right)^\times d_i - \Gamma_i d_i.$$

The motion V_U affects both bearing and range. The correction Γ_i is chosen to compensate these terms and stabilize the bearing error.

Lyapunov function and key cancellation

Choose

$$l_i(d_i, \tilde{r}_i; r_i) = \frac{r_i}{2} \|\dot{y}_i - d_i\|^2 + \frac{r_i^2}{2\alpha_i} (1 - \tilde{r}_i)^2.$$

Equivalently,

$$l_i = r_i(1 - \dot{y}_i^\top d_i) + \frac{1}{2\alpha_i} (r_i - \hat{r}_i)^2.$$

With the chosen correction terms,

$$\dot{l}_i = -k_i r_i \frac{1 - d_i^\top \dot{y}_i}{1 + d_i^\top \dot{y}_i} + (\hat{r}_i - r_i) \beta_{\tilde{r}}^\epsilon(\hat{r}_i) \leq 0.$$

Bearing convergence

Since

$$\dot{l}_i \leq 0,$$

we obtain:

- $l_i(t)$ is bounded and nonincreasing,
- $\hat{r}_i$ remains positive,
- the observer errors remain bounded.

Using standard invariance or Barbalat arguments,

$$1 - d_i^\top \dot{y}_i \rightarrow 0.$$

Since $d_i, \dot{y}_i \in \mathbb{S}^2$,

$$d_i \rightarrow \dot{y}_i.$$

Thus the landmark bearing is recovered asymptotically.

Range convergence and persistent excitation

Bearing convergence alone does not imply

$$\hat{r}_i \rightarrow r_i.$$

Range observability requires enough transverse motion.

Assume the persistence of excitation condition

$$\frac{1}{T} \int_t^{t+T} \|y_i^\times y_i^\times V_U\| d\tau \geq \mu > 0.$$

The term $y_i^\times y_i^\times V_U$ is the component of motion transverse to the line of sight.

Under this condition,

$$(d_i, \tilde{r}_i) \rightarrow (\check{y}_i, 1).$$

Hence the observer recovers bearing and range, up to the natural SLAM gauge symmetry.

Why introduce a correction term Δ ?

- The convergence result of bearings and ranges is **independent** of the choice of Δ .
- This reflects a fundamental property of SLAM: the inertial frame is **unobservable**. Different choices of Δ modify only the global rigid transform relating the reference and estimated configurations.
- Therefore, Δ cannot be determined from the SLAM error convergence itself.
- To obtain a meaningful visual-odometry map, we introduce an additional criterion: choose Δ so as to reduce the **global drift** of the estimated landmark configuration.

Drift-minimizing choice of Δ

Criterion

Assuming a static environment, choose Δ to minimize the weighted landmark velocity

$$\sum_{i=1}^n \kappa_i \|\dot{\hat{p}}_i\|^2, \quad \kappa_i > 0.$$

Writing

$$\Delta = \text{Ad}_{\hat{A}}(\Omega_{\Delta}^{\times}, V_{\Delta}),$$

the landmark kinematics yield a weighted least-squares problem in $(\Omega_{\Delta}, V_{\Delta})$, whose solution is

$$\begin{pmatrix} \Omega_{\Delta} \\ V_{\Delta} \end{pmatrix} = - \left(\sum_{i=1}^n \kappa_i \begin{pmatrix} \hat{q}_i^{\times} \hat{q}_i^{\times} & \hat{q}_i^{\times} \\ -\hat{q}_i^{\times} & I_3 \end{pmatrix} \right)^{-1} \left(\sum_{i=1}^n \kappa_i \begin{pmatrix} \hat{q}_i^{\times} \text{Ad}_{\hat{Q}_i^{\top}}(\Gamma_i) \hat{q}_i \\ \gamma_i \hat{q}_i + \text{Ad}_{\hat{Q}_i^{\top}}(\Gamma_i) \hat{q}_i \end{pmatrix} \right).$$

Interpretation: Δ is chosen to minimize the instantaneous drift of the estimated map.

Trajectories of the true and observed system

